

Eric Liu

📞 6476562258 | ✉️ haofengliu0601@gmail.com | 🏠 www.airickLeo.com | 🌐 airickLeo | 🌐 Eric

Education

University of Waterloo

Sep 2021 - Apr 2026

Candidate For Bachelor of Computer Science

Canada

Courses: Distributed Systems, Data Structures and Algorithms, Database Management, Operating Systems, OOP

Work Experience

Meta

Sep 2025 – Dec 2025

Software Engineer Intern (Upcoming)

United States

- Joining Instagram Direct Media Team

Amazon Web Services

May 2025 – Aug 2025

Software Development Engineer Intern | Rust, DynamoDB, AWS

Canada

- Built an on-demand CLI tool and real-time data emission system in **Rust**, providing live and historical observability into S3 Express object cataloging across **3,000+ buckets per region**, enabling rapid on-call investigations
- Implemented in-memory and persistence-layer checksums in the event pipeline, ensuring end-to-end data integrity
- Developed multi-level metrics and progression estimators in the CLI to surface abnormalities in live and historical views
- Engineered strongly typed I/O and automated **AWS** credential retrieval in the CLI, streamlining developer workflows

CIBC

May 2024 – Dec 2024

Software Engineer Intern | React, Spring Boot, SQL, Java, Kubernetes, Jenkins, Docker

Canada

- Delivered reporting features for an enterprise fraud management application using **React**, **Spring Boot**, and **Azure SQL**, improving fraud analysis efficiency and reducing report generation time
- Engineered **CI/CD** pipelines with **Jenkins** and **Docker** for deploying **20+ core APIs** on **OpenShift**, integrating **Blackduck** and **Fortify** for automated security scanning
- Automated database script transfers with environment setup from **Artifactory** to **Jump Server** using **GitHub Actions**
- Mitigated **95%** of security vulnerabilities in enterprise applications through security fixes and JDK version migration

Accenture Inc

Jan 2023 – Apr 2023

Software Engineer Intern | React, Express.js, JavaScript, Playwright

Canada

- Developed a JSON-driven CLI application using **TDD** to automate enterprise web-based workflows, streamlining the SDLC
- Built a single-page application for uploading, editing, executing, and visualizing JSON files for the CLI application, improving user interaction and accessibility

GrantMatch

May 2022 – Aug 2022

Data Engineer Intern | Python, SQL, Pandas, NumPy, AWS

Canada

- Led development of intuitive user interfaces for a data catalog web application using Plotly Dash and Bootstrap, leading to **30% increased user engagement** and a more efficient data discovery process
- Implemented ETL pipelines using **S3** to optimize data extraction, transformation, and storage in **PostgreSQL** database
- Developed Python scripts using **BeautifulSoup** and **Selenium** to extract **4,000+ data entries** from 10+ online platforms

Projects

UWConnect

Cloud SQL, Kotlin, Ktor, Jetpack Compose

- Led development of an Android app centralizing resources and fostering collaboration at the University of Waterloo
- Designed and implemented an MVVM architecture with Kotlin Coroutines for seamless client-server integration
- Implemented **JWT authentication** for user sign-in, action authorization, and auto re-authentication on app relaunch
- Engineered a scalable **REST API** using **Ktor** and **Cloud SQL** to support event and announcement management

Technical Skills

Languages: Rust, JavaScript/TypeScript, C/C++, Python, Kotlin, HTML, CSS, SQL, C#, Java, Bash

Frameworks: React, Express.js, Node.js, Spring Boot, Angular, Playwright, Pandas, NumPy

Tools: PostgreSQL, DynamoDB, AWS, OpenShift, Kubernetes, Docker, Jenkins, Azure, Git